

BDD Framework

Selenium · TestNG · Cucumber

■ AIM

To build a BDD (Behavior Driven Development) framework using **Selenium**, **TestNG**, and **Cucumber**.

■ PROCEDURE

1	Create Maven Project	Open IDE (Eclipse/IntelliJ) → New Maven Project
2	Add Dependencies	Add Selenium, TestNG, Cucumber in pom.xml
3	Create Page Object	Create pages package → implement LoginPage.java
4	Create Feature File	Create src/test/resources/features/ → write login.feature
5	Create Step Definitions	Create step_definitions package → implement StepDefinitions.java
6	Setup TestNG XML	Create testng.xml → configure CucumberRunner class
7	Write Selenium Code	Use WebDriver in step definitions → call LoginPage methods
8	Run Tests	Right-click testng.xml → Run As → TestNG Suite
9	Adjust Paths	Update ChromeDriver path, app URL and element locators
10	Enhance Framework	Add reporting, parallel execution, Cucumber IDE plugins

■ CODE

■ pom.xml — Dependencies

```
<dependency>
<groupId>org.seleniumhq.selenium</groupId>
<artifactId>selenium-java</artifactId>
<version>3.141.59</version>
</dependency>
<dependency>
<groupId>org.testng</groupId>
<artifactId>testng</artifactId>
<version>7.3.0</version> <scope>test</scope>
</dependency>
<dependency>
<groupId>io.cucumber</groupId>
<artifactId>cucumber-java</artifactId>
<version>7.2.10</version> <scope>test</scope>
</dependency>
<dependency>
<groupId>io.cucumber</groupId>
<artifactId>cucumber-testng</artifactId>
<version>7.2.10</version> <scope>test</scope>
</dependency>
```

■ LoginPage.java

```

package pages;

import org.openqa.selenium.*;

public class LoginPage {

    WebDriver driver;

    public LoginPage(WebDriver d) { driver = d; }

    public void open(String url) { driver.get(url); }

    public void enterUsername(String u) { driver.findElement(By.id("username")).sendKeys(u); }

    public void enterPassword(String p) { driver.findElement(By.id("password")).sendKeys(p); }

    public void clickLogin() { driver.findElement(By.id("loginButton")).click(); }

    public boolean isErrorShown() { return driver.findElement(By.id("errorMessage")).isDisplayed(); }

}

```

■ login.feature

```

Feature: Login

Scenario: Valid login
    Given User is on login page
    When User enters valid credentials
    Then User logs in successfully

Scenario: Invalid login
    Given User is on login page
    When User enters invalid credentials
    Then User sees error message

```

■ StepDefinitions.java

```

package step_definitions;

import io.cucumber.java.After; import io.cucumber.java.en.*;
import org.openqa.selenium.*; import org.openqa.selenium.chrome.ChromeDriver;
import org.testng.Assert; import pages.LoginPage;

public class StepDefinitions {

    WebDriver driver; LoginPage lp;

    @Given("User is on login page")
    public void onLoginPage() {

        System.setProperty("webdriver.chrome.driver", "path/to/chromedriver");

        driver = new ChromeDriver();

        lp = new LoginPage(driver);

        lp.open("http://yourapp.com/login"); }

    @When("User enters valid credentials")
    public void validLogin() { lp.enterUsername("admin"); lp.enterPassword("admin123"); lp.clickLogin(); }

    @When("User enters invalid credentials")
    public void invalidLogin() { lp.enterUsername("wrong"); lp.enterPassword("wrong"); lp.clickLogin(); }

    @Then("User logs in successfully")
    public void successCheck() { /* Assert welcome message */ }

```

```
@Then("User sees error message")

public void errorCheck() { Assert.assertTrue(lp.isErrorShown()); }

@After

public void tearDown() { driver.quit(); }

}
```

■ testng.xml

```
<!DOCTYPE suite SYSTEM "http://testng.org/testng-1.0.dtd">

<suite name="Suite">

<test name="BDD Tests">

<classes>

<class name="io.cucumber.testng.CucumberRunner"/>

</classes>

</test>

</suite>
```

■ RESULT

Thus, the BDD framework using **Selenium**, **TestNG**, and **Cucumber** was successfully built and executed. The login test scenarios were written in Gherkin format, step definitions were implemented using Selenium WebDriver, and tests were run successfully using the TestNG XML file.